

Nathan Potraz

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SUMMARY

Software engineer, B.S. CS (CSUSM, Dec 2024, 3.7 GPA). Ships full-stack production systems in Go and React -- REST APIs, CI/CD pipelines, Docker, GCP. Three years running a client-facing hardware business; 100+ builds delivered to professional clients. Seeking full-time entry-level roles; open to relocation.

CORE COMPETENCIES

Full-Stack Development (Go + React)

REST API Design

CI/CD Pipeline Engineering

Cloud Infrastructure (GCP/AWS)

Docker & Containerization

Systems Programming (C++/Python)

WORK EXPERIENCE

Digital Input Controller Design

June 2022 - Present

Hardware & Software Engineer - Self-Employed

- Built and operated a custom hardware engineering business end-to-end -- **100+ units delivered** to professional esports clients at \$150-\$400/unit; managed full lifecycle from client requirements and component sourcing to assembly, QA, and post-delivery support.
- Engineered performance-critical hardware for professional esports clients: circuit-board soldering, firmware optimization, and millisecond-sensitive tuning with zero tolerance for client-facing failures.

PROJECTS

Devlog

LIVE

[devlog.ntpotraz.dev](#)

- Built a production full-stack application -- **Go (net/http)** REST API backend, **TypeScript/React** + Tailwind CSS frontend -- deployed to **GCP Cloud Run** via **Docker** (Alpine). Real users, monitored in production.
- Engineered a **GitHub Actions** CI/CD pipeline (Bun + Biome + Docker + Cloud Run deploy) with automated lint, build, and deployment validation -- cutting release time from **15 minutes to under 3 minutes** and eliminating manual deploy errors.
- Integrated **Clerk** (clerk-sdk-go) for SSO authorization (GitHub + Google) on public and protected routes; implemented immutable data pattern for full edit history with zero data loss.

PIC Health & Resource Mobile App[ntpotraz.github.io/PIC-Health-App](#)

- Led a team of 5 developers using Agile sprints to architect and ship a cross-platform community hub in **React Native (Expo)** -- gathered stakeholder requirements, defined component interfaces, ran code reviews, and coordinated delivery.
- Engineered a serverless architecture at **\$0 monthly infrastructure cost** using Google Calendar API + Forms as a self-service CMS for non-technical stakeholders.

DNS Resolution Simulator[github.com/Saltair1/NetworkPythonProject](#)

- Simulated full DNS infrastructure in **Python** using UDP sockets -- modeled root, authoritative, and caching nameserver behavior with TTL expiration and **pandas** cache visualization; designed a custom byte-level protocol packing transaction IDs, query types, and flags into compact byte arrays via bitwise operators.

Japanese-English Translator[github.com/ntpotraz/cs421-translator](#)

- Built a **C++** natural language processor using DFA scanning + Recursive Descent Parsing with **100% lexical accuracy**; optimized dictionary lookups to **O(1)** via Hash Maps.
- Developed a translation engine that reordered tokens to account for syntactic differences between Japanese and English, producing grammatically correct output without a rules database.

EDUCATION

B.S. Computer Science -- California State University, San Marcos

Dec 2024

GPA: 3.7, Cum Laude | Relevant: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Security, AI, Operating Systems, Agile/SCRUM

SKILLS

Backend: Go (Golang), Python, Java, C++, SQL, PostgreSQL, MySQL, SQLite

Frontend: TypeScript, JavaScript, React, React Native, HTML/CSS

DevOps / Cloud: Docker, GitHub Actions CI/CD, GCP (Cloud Run), AWS, Linux, Git, Agile